

EC200U&EG91xU Series

QuecOpen VoLTE API

Reference Manual

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2021-12-24	Marvin NING	Creation of the document
1.0	2022-01-24	Marvin NING	First official release
1.1	2023-08-25	Marvin NING	1. Added the applicable modules EG912U-GL and EG915U series. 2. Updated the note on VoLTE feature of EC200U series module (Chapter 1). 3. Updated the log viewing information of EC200U series module (Chapter 3.2).

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1 Introduction

Quectel EC200U series and EG91xU family (EG912U-GL and EG915U series) modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see *document [1]*.

This document outlines VoLTE API and the related development example of Quectel EC200U series and EG91xU family modules in QuecOpen® solution.

NOTE

To support VoLTE feature, you need to add VoLTE when compiling the software version of EC200U series module, for example: **build_all new EC200UCN_AA EC200UCNAA VoLTE**.

2 VoLTE API

2.1. Header File

ql_api_volte.h, the header file of VoLTE API, is in the *components\ql-kernel\inc* directory. Unless otherwise specified, all the header files mentioned in this document are in this directory.

2.2. API Overview

Table 1: API Overview

Function	Description
<i>ql_volte_set()</i>	Sets VoLTE feature.
<i>ql_volte_get()</i>	Gets the status of VoLTE feature.
<i>ql_volte_vdp_set()</i>	Sets voice domain preference.
<i>ql_volte_vdp_get()</i>	Gets voice domain preference.
<i>ql_volte_usage_set()</i>	Sets VoLTE usage of the module.
<i>ql_volte_usage_get()</i>	Gets VoLTE usage of the module.

2.3. API Description

2.3.1. ql_volte_set

This function sets VoLTE feature.

- **Prototype**

```
ql_volte_errcode_e ql_volte_set(uint8_t nSim,uint8_t mode);
```

- **Parameter**

nSim:

[In] (U)SIM card in use.

0 (U)SIM 1

1 (U)SIM 2

mode:

[In] Enable or disable VoLTE feature. Default: 1.

0 Disable

1 Enable

- **Return Value**

See **Chapter 2.3.1.1** for details.

NOTE

When VoLTE feature is enabled, the range of *profile_idx* in *ql_start_data_call()* is 1–5 and manual activation of PDP context 6 or 7 is not allowed. See **document [2]** for more information about *ql_start_data_call()*.

2.3.1.1. ql_volte_errcode_e

The enumeration of VoLTE API result codes:

```
typedef enum
{
    QL_VOLTE_SUCCESS                = 0,
    QL_VOLTE_ERROR                  = 1 | (QL_COMPONENT_IMS << 16),
    QL_VOLTE_PARA_ERR,
}ql_volte_errcode_e;
```

- Member

Member	Description
QL_VOLTE_SUCCESS	Successful execution
QL_VOLTE_ERROR	Failed execution
QL_VOLTE_PARA_ERR	Parameter error

2.3.2. ql_volte_get

This function gets the status of VoLTE feature.

- Prototype

```
ql_volte_errcode_e ql_volte_get(uint8_t nSim, uint8_t *mode);
```

- Parameter

nSim:

[In] (U)SIM card in use.

0 (U)SIM 1

1 (U)SIM 2

mode:

[Out] Pointer to the status of VoLTE feature.

0 VoLTE disabled

1 VoLTE enabled

- Return Value

See **Chapter 2.3.1.1** for details.

2.3.3. ql_volte_vdp_set

This function sets voice domain preference.

- Prototype

```
ql_volte_errcode_e ql_volte_vdp_set( uint8_t nSim, uint8_t setting);
```

- **Parameter**

nSim:

[In] (U)SIM card in use.

- 0 (U)SIM 1
- 1 (U)SIM 2

setting:

[In] Voice domain preference. Default: 3.

- 1 CS Voice only
- 2 CS Voice preferred, IMS PS Voice as secondary
- 3 IMS PS Voice preferred, CS Voice as secondary
- 4 IMS PS Voice only

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.4. ql_volte_vdp_get

This function gets voice domain preference.

- **Prototype**

```
ql_volte_errcode_e ql_volte_vdp_get(uint8_t nSim, uint8_t *setting);
```

- **Parameter**

nSim:

[In] (U)SIM card in use.

- 0 (U)SIM 1
- 1 (U)SIM 2

setting:

[Out] Pointer to voice domain preference.

- 1 CS Voice only
- 2 CS Voice preferred, IMS PS Voice as secondary
- 3 IMS PS Voice preferred, and CS Voice as secondary
- 4 IMS PS Voice only

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.5. ql_volte_usage_set

This function sets VoLTE usage of the module.

- **Prototype**

```
ql_volte_errcode_e ql_volte_usage_set(uint8_t nSim, uint8_t setting);
```

- **Parameter**

nSim:

[In] (U)SIM card in use.

0 (U)SIM 1

1 (U)SIM 2

setting:

[In] VoLTE usage setting. Default: 1.

0 Voice centric

1 Data centric

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.6. ql_volte_usage_get

This function gets VoLTE usage of the module.

- **Prototype**

```
ql_volte_errcode_e ql_volte_usage_get(uint8_t nSim, uint8_t *setting);
```

- **Parameter**

nSim:

[In] (U)SIM card in use.

0 (U)SIM 1

1 (U)SIM 2

setting:

[Out] Pointer to VoLTE usage setting.

0 Voice centric

1 Data centric

- **Return Value**

See **Chapter 2.3.1.1** for details.

3 Development Example

This chapter introduces how to use the above API to develop and debug VoLTE on the application layer.

3.1. Development Example

volte_demo.c, the example file of VoLTE API, provided in the QuecOpen SDK of the module, is in the `\components\ql-application\volte` directory. The entry function is *ql_volte_app_init()* which initializes *volte_demo_task*, as shown below.

```
QLOSStatus ql_volte_app_init(void)
{
    QLOSStatus err = QL_OSI_SUCCESS;

    err = ql_rtos_task_create(&volte_task, 2048, APP_PRIORITY_NORMAL, "volte_task", volte_demo_task, NULL, 2);
    if(err != QL_OSI_SUCCESS)
    {
        QL_VOLTE_LOG("volte_demo_task created failed, ret = 0x%x", err);
    }

    return err;
}
```

As shown in the figure below, *ql_volte_app_init()* is disabled by default; if necessary, uncomment it in `components\quectel\ql_open\ql_examples\init\ql_init.c` file.

```
#ifdef QL_APP_FEATURE_VOLTE
    ql_volte_app_init();
#endif
```

3.2. Feature Debugging

The module's VoLTE feature can be debugged on Quectel LTE OPEN EVB.

Download the compiled firmware to the module and connect the USB port of the LTE OPEN EVB to a PC with a USB cable. The USB AP Log Port mainly displays the system debugging information. You can view information of the example through USB AP Log Port after capturing the log with cooltools. For details about log capture, see **document [3]**.

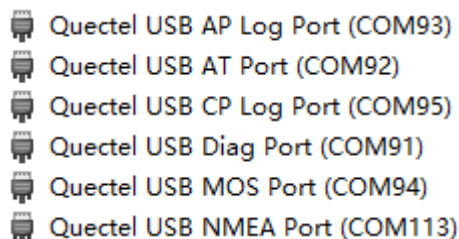


Figure 1: Ports in Device Manager

After *volte_demo_task* starts, it automatically enables VoLTE feature and sets data centric and IMS PS Voice preferred, as shown below.

```
void volte_demo_task(void * param)
{
    ql_volte_errcode_e err = QL_VOLTE_SUCCESS;
    uint8_t nSim = 0;

    //China Telecom 4G/5G SIM card must enable VoLTE before making a call
    err = ql_volte_set(nSim, 1);
    if(err != QL_VOLTE_SUCCESS){
        QL_VOLTE_LOG("ql_voice_call_volte_set FAIL");
    }else {
        QL_VOLTE_LOG("ql_voice_call_volte_set OK");
    }

    //data centric
    err = ql_volte_usage_set(nSim, 1);
    if(err != QL_VOLTE_SUCCESS){
        QL_VOLTE_LOG("data centric set FAIL,err = %d",err);
    }else{
        QL_VOLTE_LOG("data centric set OK");
    }

    //set IMS PS Voice preferred, CS Voice as secondary
    err = ql_volte_vdp_set(nSim, 4);
    if(err != QL_VOLTE_SUCCESS){
        QL_VOLTE_LOG("set IMS PS Voice preferred FAIL,err=%d",err);
    }else{
        QL_VOLTE_LOG("set IMS PS Voice preferred OK");
    }

    ql_rtos_task_delete(NULL);
} « end volte_demo_task »
```

4 Appendix References

Table 2: Related Documents

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen_CSDK_Quick_Start_Guide
[2] Quectel_EC200U&EG91xU_Series_QuecOpen_Data_Call_API_Reference_Manual
[3] Quectel_EC200U&EG91xU_Series_QuecOpen_Log_Capture_Guide

Table 3: Terms and Abbreviations

Abbreviation	Description
(U)SIM	(Universal) Subscriber Identity Module
API	Application Programming Interface
CS	Circuit Switching
EVB	Evaluation Board
IMS	IP Multimedia Subsystem
IoT	Internet of Things
LTE	Long Term Evolution
PC	Personal Computer
PS	Packet Switch
RTOS	Real-Time Operating System
SDK	Software Development Kit
USB	Universal Serial Bus
VoLTE	Voice (voice calls) over LTE